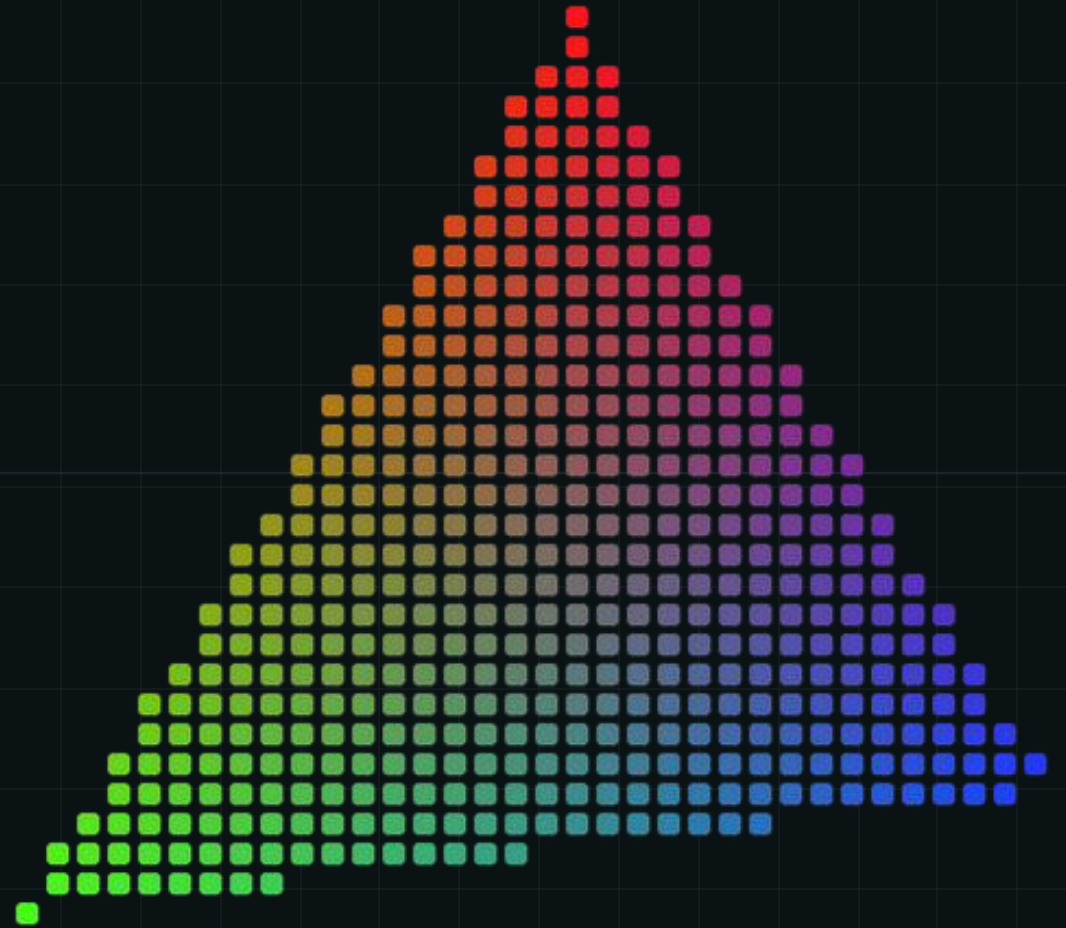
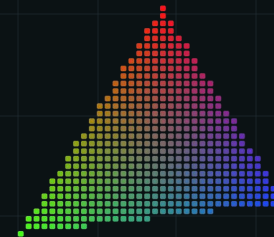


POTATO GPU

POTATO Optimizes Texture And Throughput Operations



// one pipeline. every pixel. real time.



THE TEAM



Lukas Schwarzmeier

393158 CE



David Schröder

Schülerstudent



Silvan Josten

410678 EE

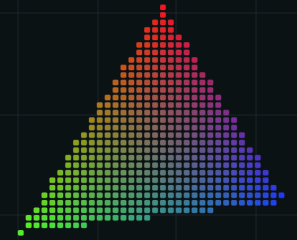


Ruslan Kharasov

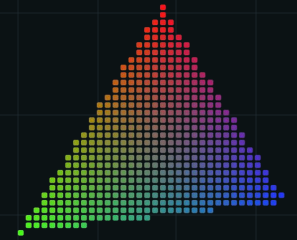
486591 TI

THE DREAM

A **high-resolution** video controller for **interactive 3D environments**



// manifesting a better future



THE DREAM

// brilliant colors & razor-sharp — right in the thick of it

A **high-resolution** video controller for **interactive 3D environments**

Full HD

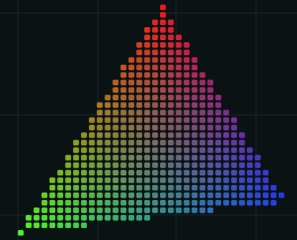
60Hz

24b color

Triangle mesh

High-resolution textures

Responsive input



THE DREAM

A **high-resolution** video controller for **interactive 3D environments**

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24b color

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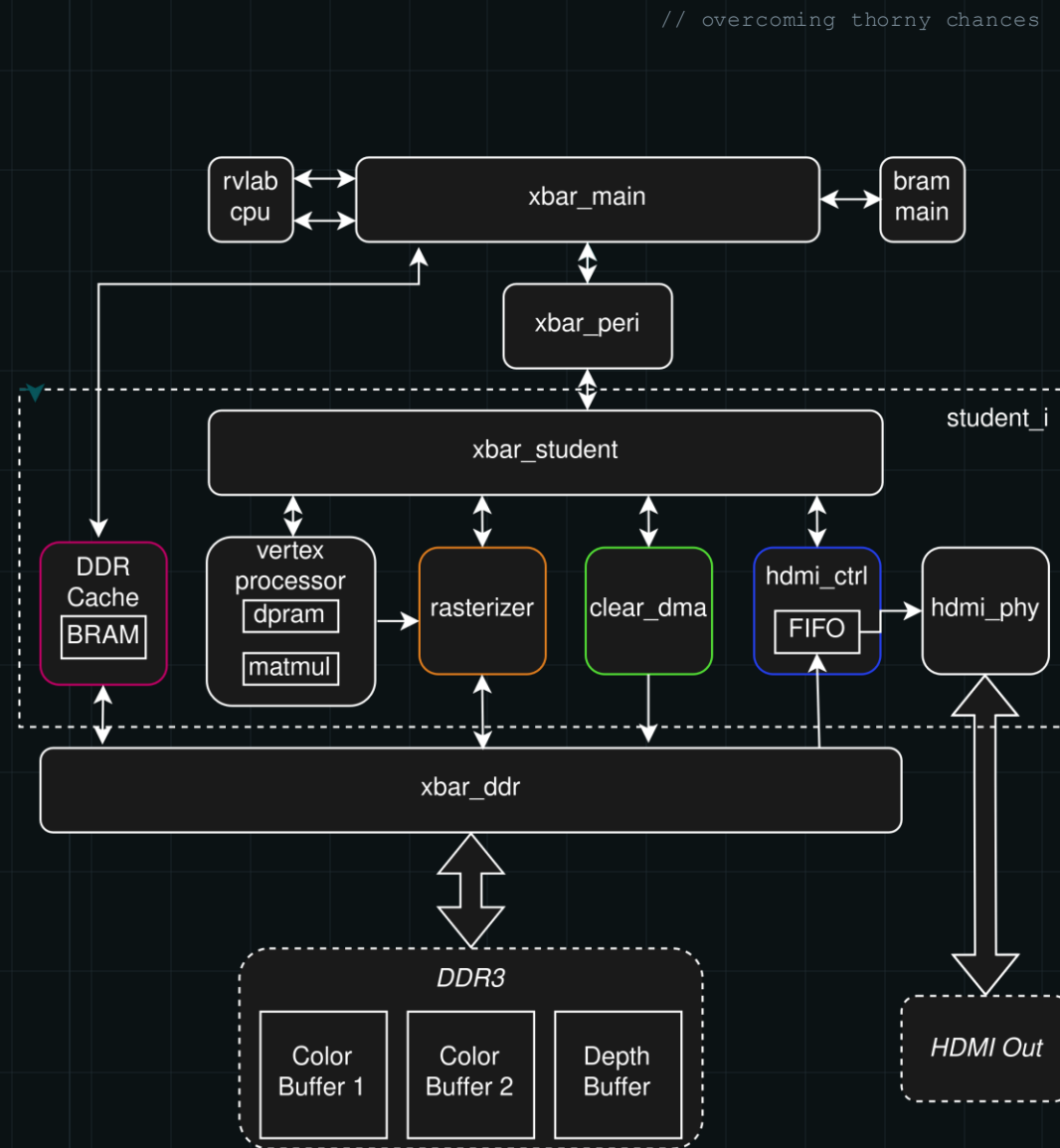
6MB frame buffer must be stored in DRAM!

➡ Primary challenge: **resource management**

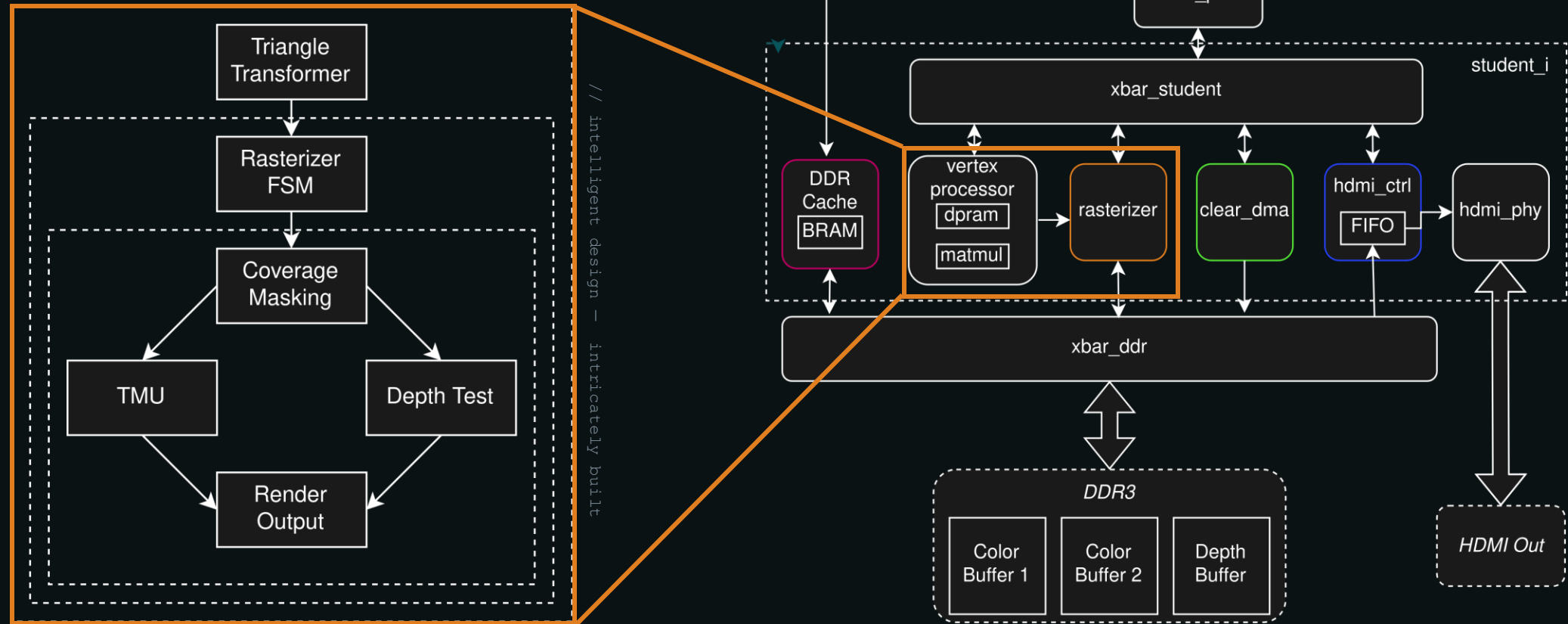
THE CHALLENGE

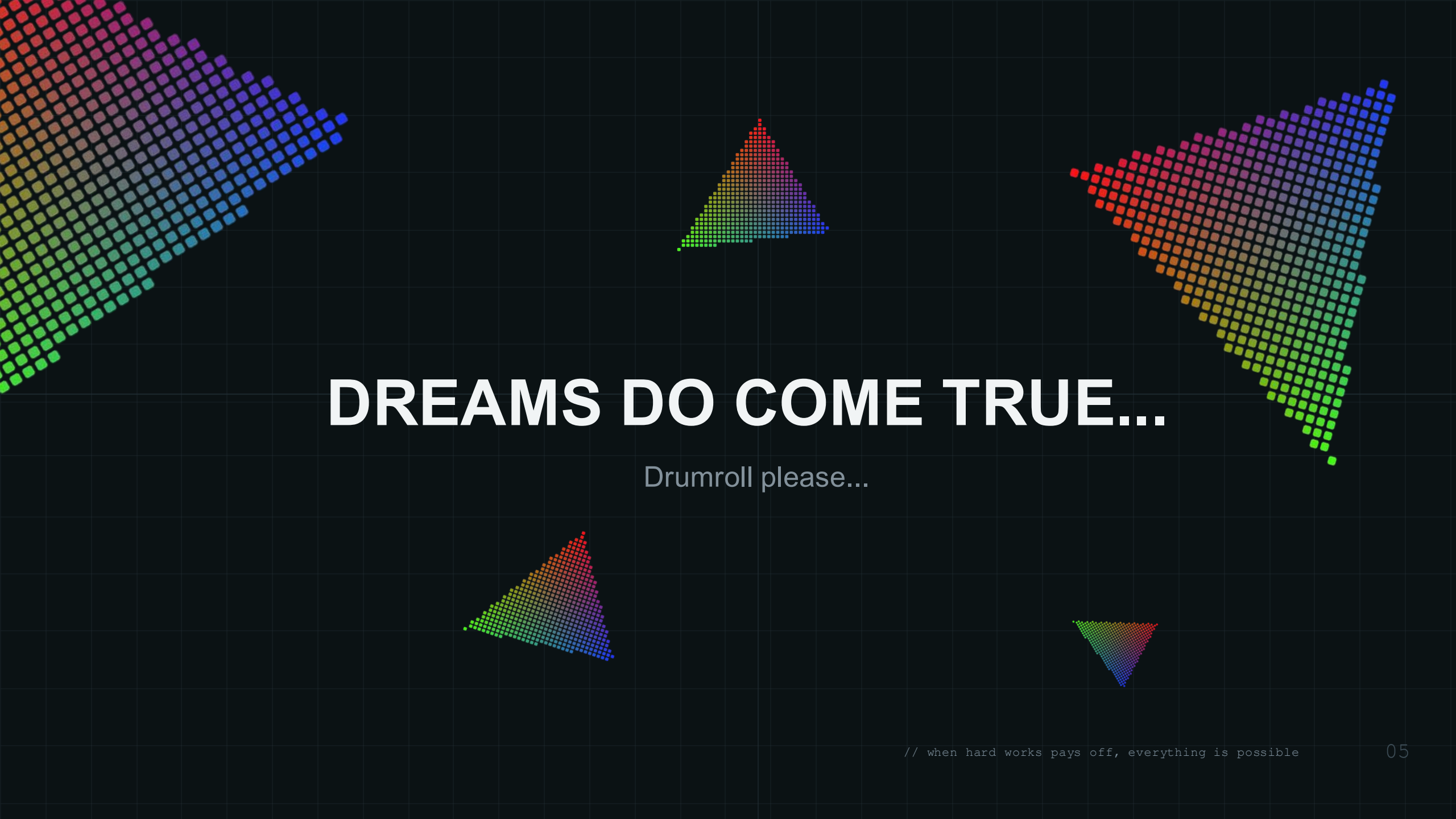
Many consumers of DDR3 bandwidth:

- **Framebuffer read** for HDMI output
- Frame + depth **buffer clear** every frame
- **Depth testing** transactions
- Mass texture/mesh **storage** (CPU)



THE ARCHITECTURE



The background features a dark gray grid. Four colorful, pixelated triangles are positioned around the central text. One large triangle is in the top-left corner, pointing towards the center. A smaller one is at the top-center, pointing down. Another large one is in the top-right corner, pointing towards the center. A smaller one is at the bottom-left, pointing up. A small one is at the bottom-right, pointing up. All triangles have a rainbow gradient from red at the top to blue at the bottom.

DREAMS DO COME TRUE...

Drumroll please...

// when hard works pays off, everything is possible